

# Kinematics-Driven Reconstruction of Microscale Strain Localization in 316L Stainless Steel

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## ABSTRACT

This study proposes a kinematics-driven experimental–numerical framework for reconstructing microscale strain localization in 316L stainless steel. Locally effective elastoplastic descriptors are identified pixel-wise from digital image correlation (DIC) strain histories and mapped onto a regular two-dimensional finite element mesh at DIC pixel resolution. These descriptors capture the spatial organization of the measured kinematic response under the imposed loading conditions; they are not intrinsic grain-scale material properties, and their values depend on the loading path, DIC spatial resolution, and constitutive assumptions. Boundary conditions prescribed from measured displacement fields drive the finite element model to enforce mechanical equilibrium and produce a physically admissible reconstructed strain field.

The numerical simulations reproduce the main displacement and strain localization features observed experimentally. The displacement RMSE is  $1.32 \times 10^{-2} \%$ , and the von Mises equivalent strain RMSE is 0.361% over the  $3600 \times 3100$ -pixel region of interest. The framework is computationally lightweight relative to crystal plasticity approaches and requires no microstructural input, offering a practical tool for kinematics-driven microscale strain reconstruction.

## 1. Introduction

Fatigue and failure in metallic materials are strongly influenced by localized plastic deformation at the grain scale, which acts as a precursor to crack initiation [1–4]. In heterogeneous metals, microstructural features such as grain boundaries, inclusions, and phase interfaces govern the development of strain localization and the overall mechanical response under monotonic or cyclic loading [5, 6]. Translating locally measured kinematic heterogeneity into mechanically consistent constitutive fields at this scale remains an open challenge.

Digital image correlation (DIC) has become the standard technique for studying strain heterogeneity in polycrystalline alloys, providing high-resolution full-field displacement and strain measurements down to the sub-granular scale [7, 8]. In 316L austenitic stainless steel, DIC has been used to characterize strain localization bands, to study the influence of grain orientation and grain boundary character on local deformation, and to provide experimental data for the calibration of numerical models [9, 10]. However, DIC-derived strain fields do not inherently satisfy mechanical equilibrium, and the numerical differentiation of measured displacement fields amplifies measurement noise [11]. Their direct use as constitutive inputs therefore requires a framework that explicitly couples experimental observation with mechanical admissibility.

To enforce equilibrium and recover stress fields, DIC data are commonly coupled with finite element models. Crystal plasticity finite element methods (CPFEM) offer a physically grounded framework for this purpose and have been applied to 316L stainless steel to reproduce grain-scale deformation patterns when microstructural information is available [9, 12–14]. However, CPFEM requires grain orientation data, single-crystal elastic constants, and slip-system parameters, and its computational cost limits applicability to large domains or systematic studies [15]. This motivates lighter alternatives that retain mechanical admissibility without requiring detailed microstructural input.

The constitutive framework adopted here is the Ludwik–Hollomon power-law relation [16, 17]. This choice reflects a deliberate simplification: the objective is not to construct the richest possible microstructural description, but to identify the minimum constitutive complexity needed to translate experimentally observed kinematic heterogeneity into a locally structured numerical response. The three-parameter law captures the essential features of elastoplastic

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hardening and is well suited for integration with full-field kinematic data. For 316L stainless steel, which exhibits smooth strain hardening and limited anisotropy in the rolled condition, this formulation provides a physically reasonable and computationally efficient description [18, 19].

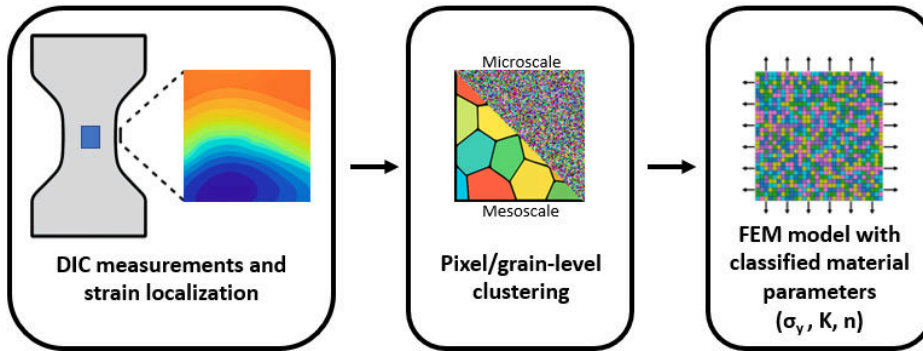
Data-driven and machine-learning approaches have also been explored for element-scale identification of elastoplastic parameters from strain fields [20–29]. At the microscale, however, strong local heterogeneities, measurement noise, and the absence of explicit physical constraints can yield non-physical parameter estimates and complicate the interpretation of identified quantities.

The present work builds on the kinematics-driven identification framework of Hu et al. [30], in which effective elastoplastic descriptors are extracted pixel-wise from DIC strain histories and a scaling relation redistributes macroscopic hardening behavior according to the locally observed kinematic response. These descriptors are not intrinsic grain-scale material properties — their values depend on the loading path, the DIC spatial resolution, and the chosen constitutive form — but effective local quantities whose primary role is to reproduce the spatial organization of deformation. Starting from these identified fields, a regular two-dimensional finite element mesh is constructed at pixel resolution, experimentally measured displacement fields are prescribed as boundary conditions, and the model is solved to produce a mechanically admissible reconstructed strain field. The central question is whether this chain — macro law, local heterogenization, numerical reconstruction — is sufficient to reproduce the dominant patterns of microscale strain localization without resorting to crystal plasticity.

The paper is organized as follows. Section 2 describes the material and constitutive assumptions, DIC data handling and mesh construction, local parameter identification, finite element implementation, and macroscopic stress–strain reconstruction. Section 3 presents the comparison between experimental and numerical fields and discusses the consistency of the reconstructed macroscopic response. Section 4 summarizes the main findings and the conditions under which this level of description is sufficient.

## 2. Methodology and Modeling Framework

The methodology follows a kinematics-driven hierarchical framework in which experimentally observed displacement fields are used to infer locally effective constitutive parameters, which are then mapped into a finite element model to reconstruct a mechanically admissible strain response. The overall workflow is illustrated in Fig. 1.



**Figure 1:** Overall workflow: (i) tensile test and DIC to obtain displacement and strain fields; (ii) pixel-wise identification of locally effective elastoplastic parameters; (iii) 2D finite element mesh with mapped elastoplastic properties and prescribed displacement boundary conditions.

### 2.1. Material and constitutive assumptions

The material investigated is a rolled 316L austenitic stainless steel. Macroscopic tensile testing was performed on an electromechanical Instron 5580 testing machine under displacement-controlled loading. The specimen was loaded up to an engineering strain of approximately 0.25% using a crosshead speed of  $1 \text{ mm min}^{-1}$ , corresponding to a strain rate of about  $2 \times 10^{-3} \text{ s}^{-1}$ . Additional details on specimen geometry, experimental setup, and measurement procedure are given in Hu et al. [30].

81 The constitutive behavior is described by the Ludwik–Hollomon power-law relation, selected for its ability to  
 82 capture elastoplastic hardening with a minimal number of parameters:

$$\sigma = \sigma_y + K \cdot \varepsilon_p^n, \quad (1)$$

83 where  $\sigma$  is the equivalent stress,  $\sigma_y$  the yield stress,  $K$  the hardening coefficient,  $\varepsilon_p$  the equivalent plastic strain, and  
 84  $n$  the strain-hardening exponent. The use of a simplified isotropic law intentionally limits constitutive complexity,  
 85 allowing the contribution of the experimental–numerical coupling to be assessed independently from more elaborate  
 86 mechanisms such as latent hardening or slip-system interactions. The macroscopic parameters  $K$  and  $n$  are identified  
 87 from the tensile test data and reported in Table 1; they are held spatially constant in the finite element model.

88 Fig. 2 shows the macroscopic equivalent stress–strain response obtained from the tensile test, from which the  
 89 parameters in Table 1 are identified.

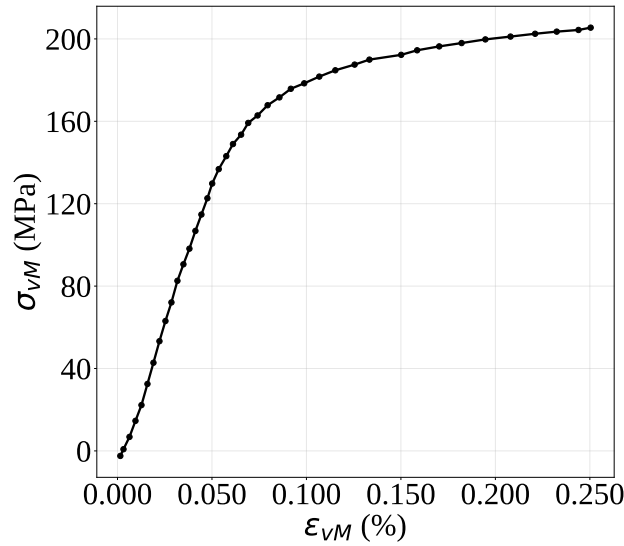


Figure 2: Macroscopic stress–strain curve.

Table 1

Material constants determined at the macroscale

| Parameter  | Value   |
|------------|---------|
| $E$        | 205 GPa |
| $\nu$      | 0.30    |
| $n$        | 0.245   |
| $\sigma_y$ | 124 MPa |
| $K$        | 380 MPa |

90 Parameters identified from DIC at the pixel scale are treated as locally effective descriptors of the measured surface  
 91 kinematics, not as intrinsic constitutive properties of individual microstructural domains. Their values are conditional  
 92 on the loading path, spatial resolution, and DIC filtering, and should be interpreted as reconstruction-oriented quantities  
 93 rather than as grain-scale material constants.

94 A plane stress assumption is adopted since the analysis is based on surface DIC measurements from a thin tensile  
 95 specimen of 2 mm thickness. The experimental observation window initially covers a  $7 \times 10 \text{ mm}^2$  surface region of  
 96 interest (ROI), from which a cropped region of  $3600 \times 3100$  pixels is retained for the finite element reconstruction.  
 97 With a pixel size of  $1.84 \mu\text{m}$ , this corresponds to a physical domain of  $6.624 \times 5.704 \text{ mm}^2$ , which remains larger than  
 98 the specimen thickness and is therefore consistent with the plane stress approximation.

Under plane stress, the out-of-plane normal stress vanishes ( $\sigma_{zz} = 0$ ), while the corresponding strain component is given by:

$$\epsilon_{zz} = -\frac{\nu}{1-\nu} (\epsilon_{xx} + \epsilon_{yy}). \quad (2)$$

The scalar strain measure used throughout is the von Mises equivalent strain derived from the three-dimensional deviatoric invariant. Under the plane stress assumption, this reduces to:

$$\epsilon_{\text{vM}}^{\text{ps}} = \frac{2}{3(1-\nu)} \sqrt{(1-\nu)^2 (\epsilon_{xx}^2 + \epsilon_{yy}^2 - \epsilon_{xx}\epsilon_{yy} + 3\epsilon_{xy}^2) + \nu (\epsilon_{xx} + \epsilon_{yy})^2}. \quad (3)$$

Here  $\nu$  is Poisson's ratio and  $\epsilon_{xy}$  is the tensorial shear strain. The detailed derivation from the three-dimensional deviatoric invariant is provided in Appendix B. Although the macroscopic loading is nominally uniaxial, microscale kinematics exhibit heterogeneous multi-axial strain components and local rotations, making an invariant scalar measure necessary for consistent field comparisons.

To preserve first-order consistency between locally identified parameters and the global hardening response, a scaling relation links the local slope of the plasticity curve to the macroscopic hardening coefficient:

$$\frac{\alpha^l}{\alpha^g} = \frac{K^g}{K^l}, \quad (4)$$

where  $\alpha^l$  is the local slope of the pixel-level plasticity curve,  $\alpha^g$  is the slope of the macroscopic plasticity curve,  $K^l$  is the local Ludwik hardening coefficient, and  $K^g$  is its macroscopic hardening coefficient. The identification procedure following from this relation is described in Hu et al. [30]. Variability in the identified parameters reflects both genuine material heterogeneity and uncertainty in the DIC strain fields, particularly in regions with limited plastic activity.

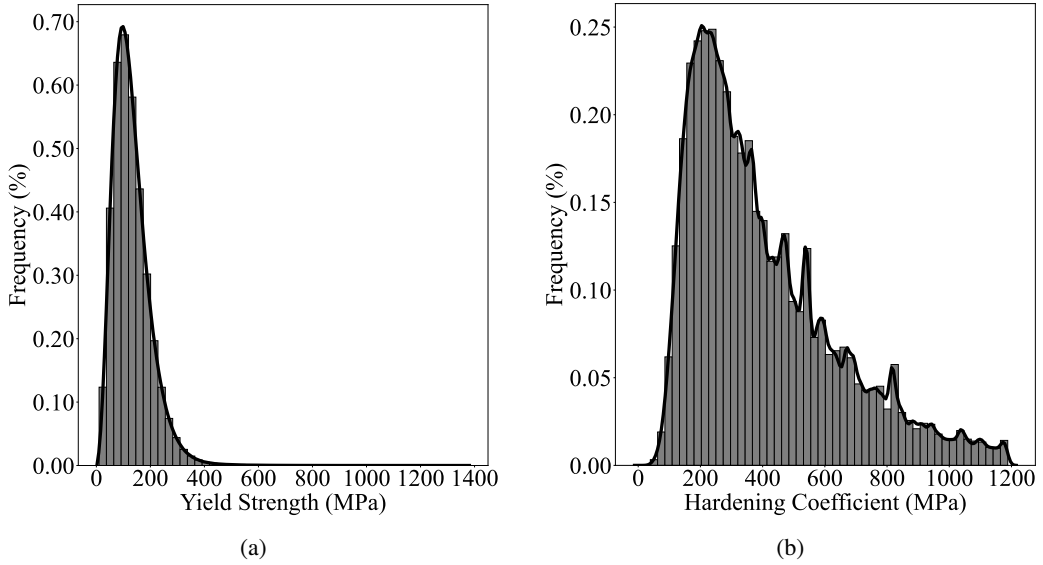
## 2.2. DIC data handling and mesh construction

Full-field displacement measurements are obtained using a Python-based DIC pipeline in which inter-frame motion is estimated by the Dense Inverse Search optical-flow (DISFlow) algorithm implemented in OpenCV [31]. Displacement fields serve as kinematic inputs, from which strain fields are computed by numerical differentiation under the plane stress assumption.

The DISFlow parameters were set to  $\alpha = 100$ ,  $\delta = 1$ ,  $\gamma = 0$ ,  $\epsilon = 0.002$ , with a maximum of 30 gradient-descent iterations. The parameter  $\alpha$  controls spatial smoothness regularization,  $\delta$  weights the brightness-constancy constraint, and  $\gamma$  controls the gradient-constancy term.

Displacement accuracy is influenced by local image texture, contrast, and gray-level distribution, which affect the robustness of the correlation procedure [32–34]. Spatial variations in speckle quality introduce heterogeneous measurement noise, which is further amplified during strain computation — particularly for small-magnitude components. The DIC-derived fields are therefore treated as experimentally consistent but noise-affected inputs that constitute the observed kinematic basis of the identification procedure.

The DIC data are mapped onto a regular two-dimensional finite element mesh in which each element corresponds to one experimental pixel. The ROI contains  $3600 \times 3100$  pixels, giving approximately 11.16 million elements over a physical domain of  $6.624 \times 5.704$  mm<sup>2</sup>. The calculated material properties reserve measurement noise, which result in nonphysical yield stresses and hardening coefficients at the lower and upper extremes (Fig. 3).



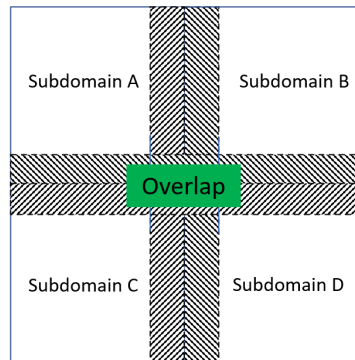
**Figure 3:** Frequency distribution of estimated a) yield strength and (b) hardening parameters (MPa) at the microscale with 50 bins

### 2.3. Finite element implementation

All simulations are carried out in Abaqus. Elastic properties ( $E$ ,  $\nu$ ) are assumed homogeneous over the entire domain. In the elastoplastic simulations, the material behavior at each element is defined through a stress–plastic strain table generated from Eq. (1) over a plastic strain range  $0 \leq \epsilon_p \leq 0.2$ , discretized using 1000 uniformly spaced points with a minimum increment of  $10^{-6}$ . Experimentally measured displacement fields are prescribed as boundary conditions on the subdomain edges, ensuring kinematic consistency between the simulation and the experiment.

### 2.4. Domain partitioning and stitching

Running a single finite element simulation over all 11.16 million elements is computationally expensive (Table 2): the memory and wall-time demand of a monolithic solve at this resolution far exceed what a single job can accommodate. The domain is therefore partitioned into a regular (depending on the padding) grid of subdomains, each solved independently and subsequently stitched into the global field, as illustrated in Fig. 4.



**Figure 4:** Schematic of the overlapping partitioning scheme (shown for a  $2 \times 2$  grid). Each subdomain (A–D) is extended into its neighbors' territory; the hatched region is shared, overlapping material solved redundantly by each of the adjacent subdomains. Only the non-overlapped core of each subdomain is retained and stitched into the global field.

In a direct, non-overlapping partitioning, each subdomain is solved with the experimentally measured displacement field prescribed directly at its own boundary. This kinematic constraint is imposed with no information about the

material beyond that boundary: the neighboring subdomain is simply absent from the model, therefore, the boundary behaves as a rigid wall for the equilibrium solution. Adjacent subdomains consequently converge to slightly different stress and strain states along their shared interface, producing sharp discontinuities — visible as artifact lines — once the subdomains are stitched into the global field.

This is a well-known difficulty in non-overlapping domain decomposition, and its resolution follows the overlapping (Schwarz) family of domain decomposition methods [35, 36]: each subdomain is extended by an overlap region into its neighbors' territory before solving, and only the non-overlapped core is retained once the subdomain solutions are assembled. The overlap gives the artificially imposed boundary condition room to relax toward the true equilibrium state before reaching the region of interest, consistent with a discrete Saint-Venant-type decay of boundary influence with distance [37]. In the present work this overlap is implemented as the padding described below, and the 100 partition results (Table 2) are assembled into the global field by stitching the non-overlapped cores.

To determine the required padding width, a convergence study was conducted on a reduced domain concerning the computation cost efficiency. The boundary gradient error (BGE) metric quantifies the ratio of strain discontinuity at partition interfaces to the field mean; it converges to unity as artifacts vanish. The study shows that for the von Mises equivalent strain, BGE reaches its converged value at a padding of approximately 150 elements, corresponding to roughly 21% of the core width. This value is used in all production simulations. The stitching artifact lines visible at subdomain interfaces in the strain difference map (Fig. 9) reflect residual mismatches from this finite padding width.

The number of subdomains represents a weak compromise between per-job memory demand and wall-clock time. Table 2 summarizes the computational cost measured for the two partitioning schemes evaluated (25 and 100 subdomains, both with a padding of 150 elements), alongside an estimate for a single monolithic solve of the full domain. However, despite the multiple partitioning attempts, the final choice of the partition numbers for the study remains to be deliberate, driven by the memory ceiling of the available hardware rather than a systematic comparison against the residual stitching artifact level.

**Table 2**

Computational cost per job as a function of partitioning scheme. Time and memory are per individual Abaqus/Standard job (input-file processing plus solve, excluding queue wait); jobs within a scheme run independently and in parallel time.

| Computation scheme                             | Time per job              | Peak memory (MaxRSS) each   |
|------------------------------------------------|---------------------------|-----------------------------|
| Full domain (1 job, 11.16M elements)           | ~5.5 days                 | ~430 GB                     |
| 25 partitions without padding (446k elem./job) | ~1.5–2 h                  | ~24 GB                      |
| 25 partitions with padding (~600k elem./job)   | ~2–4.7 h                  | 32.0–38.4 GB (mean 36.5 GB) |
| 100 partitions with padding (~350k elem./job)  | 0.99–3.70 h (mean 1.77 h) | 12.2–20.8 GB (mean 18.7 GB) |
| Output post-processing (per job)               | 0.2–10 h (per job)        | 1 GB–13GB                   |

## 2.5. Macroscopic stress–strain reconstruction

Macroscopic stress–strain responses are reconstructed from both DIC and finite element fields using spatially averaged quantities. An equivalent yield strain is defined as

$$\epsilon_y = \frac{\sigma_y}{3G}, \quad (5)$$

and the equivalent plastic strain as

$$\epsilon_p = \max(\epsilon_{vM}^{ps} - \epsilon_y, 0). \quad (6)$$

The equivalent stress is then obtained from the piecewise constitutive relation:

$$\sigma_{eq} = \begin{cases} 3G \epsilon_{vM}^{ps}, & \epsilon_{vM}^{ps} \leq \epsilon_y, \\ \sigma_y + K (\epsilon_p)^n, & \epsilon_{vM}^{ps} > \epsilon_y. \end{cases} \quad (7)$$

This formulation provides a consistent scalar framework for comparing experimentally derived and numerically computed responses based on the same invariant strain measure.

### 3. Results and discussion

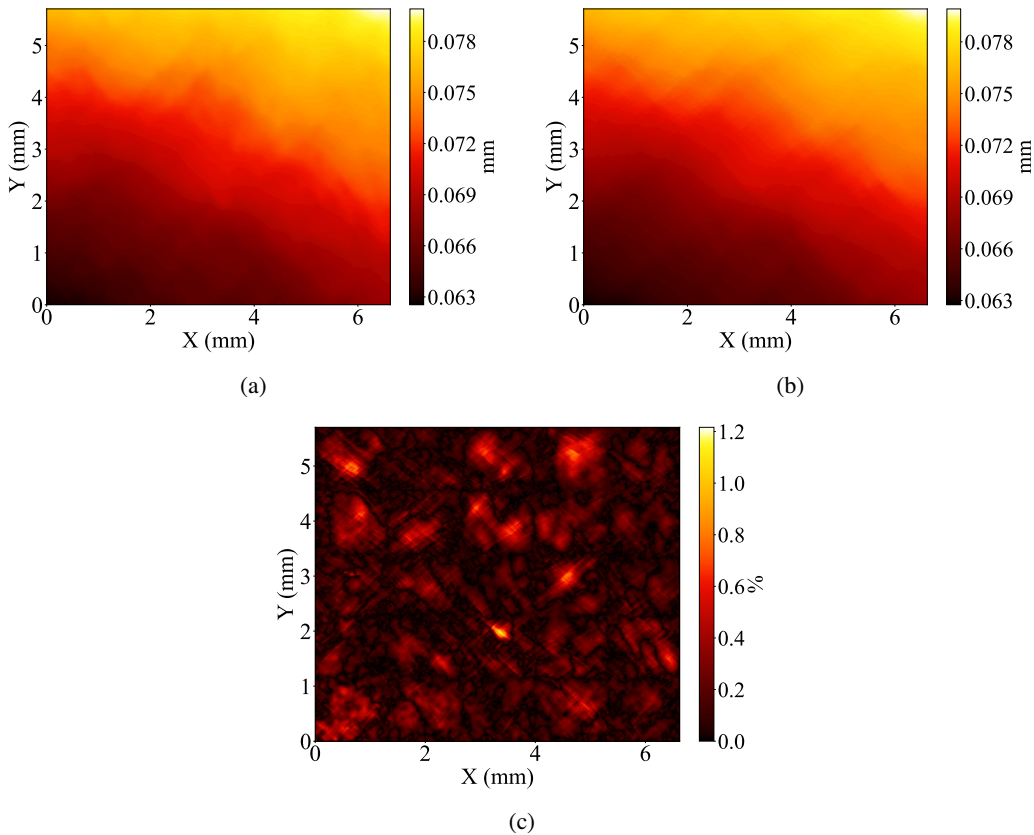
All comparisons are performed from displacement-derived strain fields computed with NumPy's gradient function, consistent with the numerical differentiation applied to DIC data and avoiding interpolation-related variations introduced by Abaqus visualization.

#### 3.1. Reconstruction of displacement and strain fields at the microscale

Unless stated otherwise, the results below are reconstructed with 25 direct, non-overlapping partitions (no padding); the effect of the 100-subdomain, pad-150 overlapping scheme is examined separately.

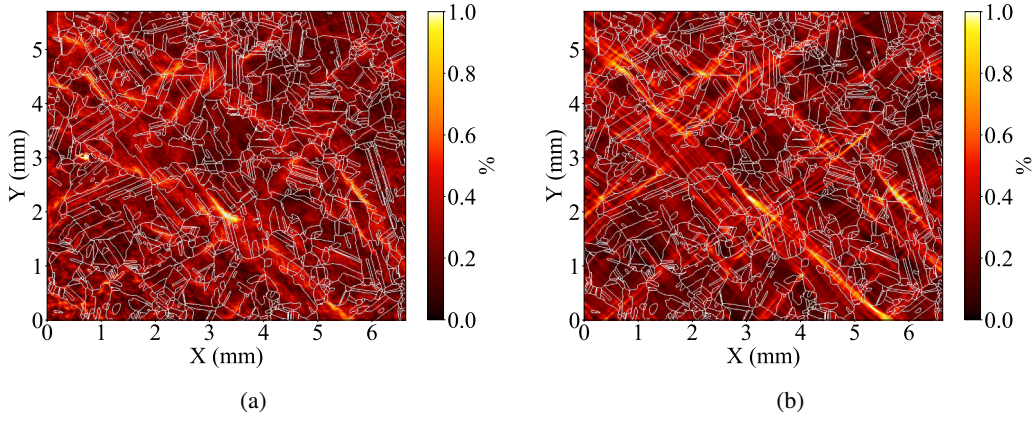
The stitched numerical displacement field closely follows the experimental pattern. The difference field in Fig. 5c shows near-zero values over most of the domain, with localized deviations near the partition interfaces attributable to stitching artifacts and local boundary condition mismatches.

Fig. 6 compares the experimental and numerical von Mises strain fields, saturated at 1% for visualization. The numerical field reproduces the main localization bands observed experimentally, though the strain distribution is smoother and more spatially diffuse. The experimental field shows sharper, more localized strain concentrations — partly reflecting genuine material heterogeneity and partly high-frequency kinematic noise amplified by the DIC differentiation step. Equilibrium enforcement in the finite element model redistributes these local extrema over neighboring elements, producing broader but mechanically consistent localization structures.



**Figure 5:** Comparison of the vertical displacement field  $U_2$ : (a) experimental field, (b) numerical field, and (c) relative difference/error field. The tensile loading direction corresponds to the  $Y$ -axis.

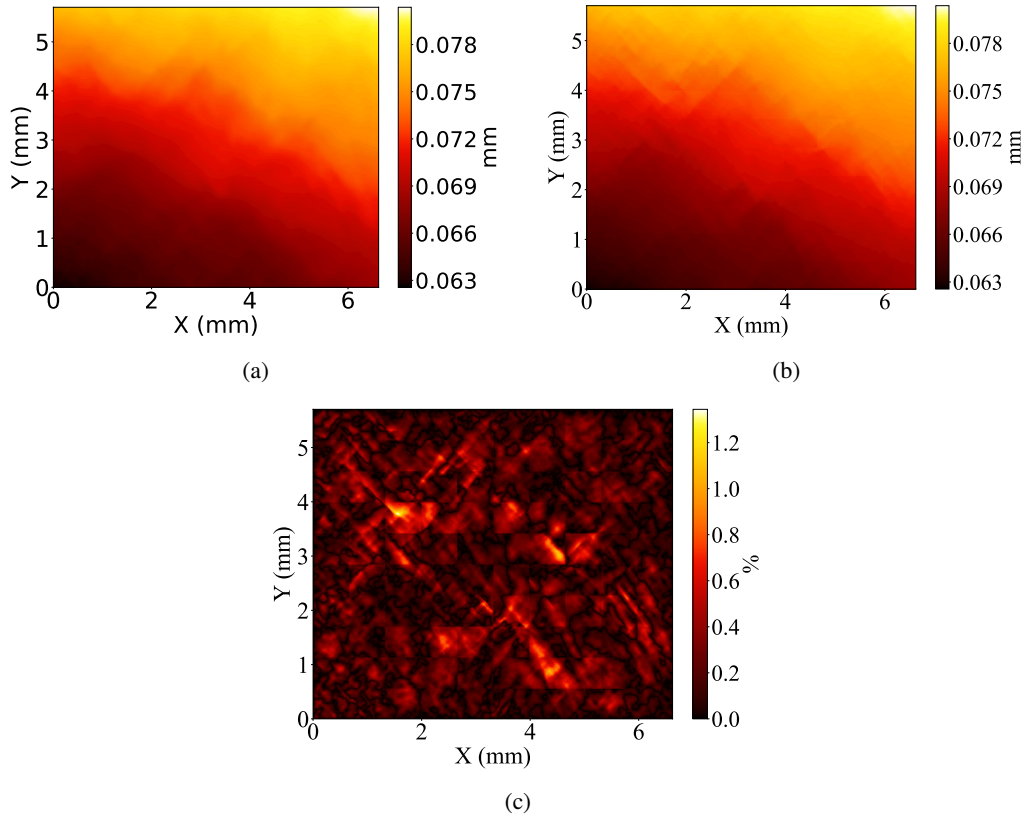




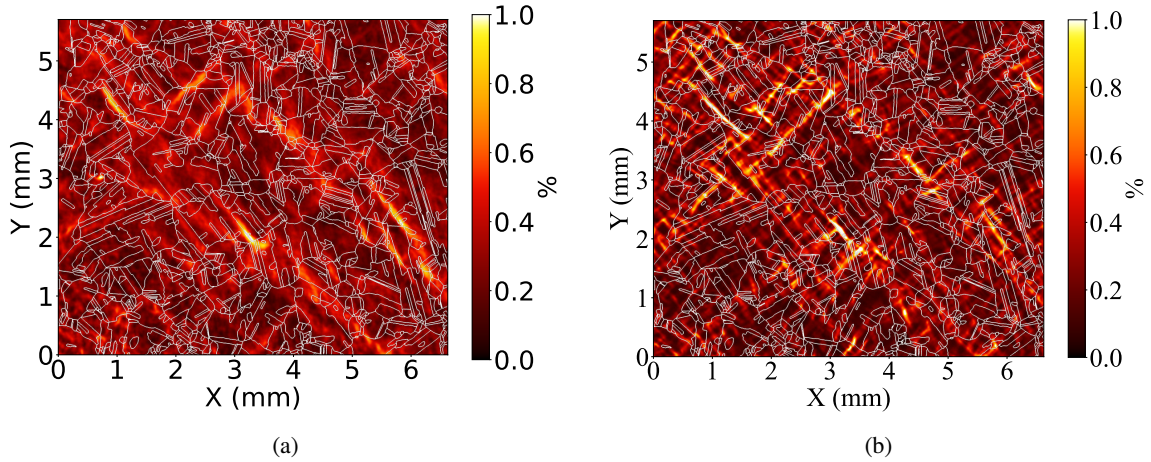
**Figure 6:** Experimental (a) and numerical (b) von Mises strain fields with values saturated at 1% for visualization.

189 Fig. 7 and Fig. 8 reproduce the displacement and von Mises strain comparison as Fig. 5 and Fig. 6, but using the  
 190 100-subdomain, pad-150 overlapping partitioning scheme (Table 2) instead of the 25 direct, non-overlapping partitions  
 191 used above. The displacement field itself (Fig. 7b) is in agreement with the 25-partition case, but its difference map to  
 192 the experimental measurement (Fig. 7c) shows a faint grid pattern at the  $10 \times 10$  partition boundaries, spaced at the core  
 193 partition size (0.66 mm) – the residual Schwarz-overlap mismatch quantified in Section 2.4. This pattern is not visible  
 194 in the von Mises strain maps (Fig. 8), since those are computed from Abaqus’s own per-element strain output rather  
 195 than by differentiating the stitched displacement field, so the seam mismatch is not re-amplified through numerical  
 196 differentiation.





**Figure 7:** Comparison of the vertical displacement field  $U_2$  with the 100-subdomain, pad-150 overlapping partitioning scheme (compare Fig. 5): (a) experimental field, (b) numerical field, and (c) relative difference/error field.



**Figure 8:** Experimental (a) and numerical (b) von Mises strain fields with values saturated at 1% for visualization, with the 100-subdomain, pad-150 overlapping partitioning scheme (compare Fig. 6).

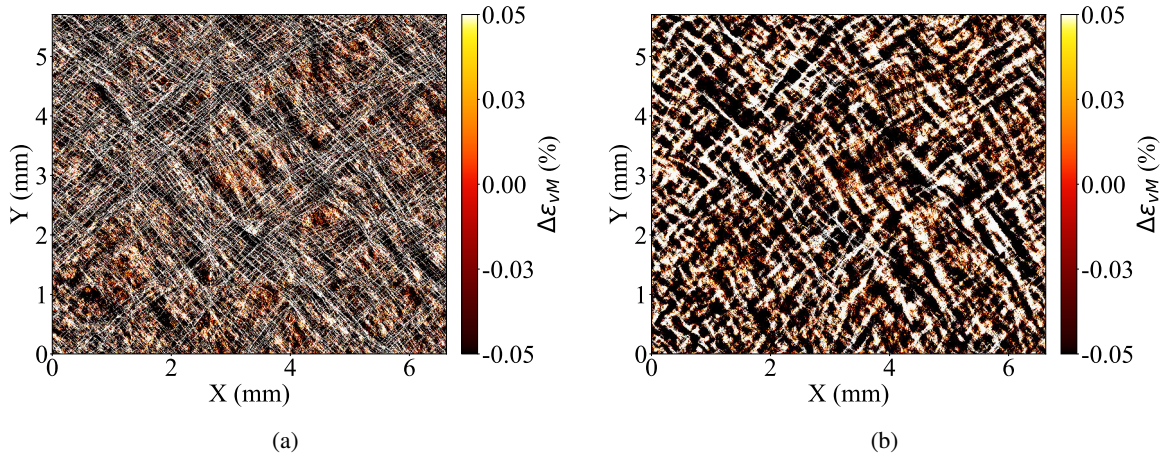
### 3.2. Quantitative field comparison between experimental and numerical results

Fig. 9 shows the von Mises strain difference map between the stitched numerical and experimental fields. The RMSE values reported below are computed over all 11,160,000 pixels of the  $3600 \times 3100$  cropped ROI.

For the displacement field  $U_2$ , the RMSE is  $1.32 \times 10^{-2} \%$ , confirming that the numerical model reproduces the measured displacement field with high consistency in both magnitude and spatial distribution.

For the von Mises equivalent strain, the RMSE is 0.361%, compared with a mean experimental equivalent strain of approximately 0.25% in the ROI. The elevated RMSE arises primarily from differences in local strain amplitudes rather than from a mismatch in the overall spatial organization of deformation, and integrates contributions from stitching interface regions where local differences exceed the visualization range. The mean absolute difference over the field is 0.039% equivalent strain.

Fig. 9b shows the same signed difference map for the 100-subdomain, pad-150 overlapping scheme (Fig. 8). The RMSE is 0.220% and the mean absolute difference is 0.156%, both lower than the 25-partition values above, but the map is visibly noisier: the finer  $10 \times 10$  partitioning increases the relative weight of per-partition boundary and DIC pixel-level noise in the signed difference, so the saturation at  $\pm 0.05\%$  is reached over a larger fraction of the field despite the lower RMSE.



**Figure 9:** Signed difference map between the experimental and numerical von Mises equivalent strain fields at the microscale, defined as  $\Delta\epsilon_{vM} = \epsilon_{vM}^{num} - \epsilon_{vM}^{exp}$ , shown on a common color scale saturated at  $\pm 0.05\%$ : (a) 25 direct, non-overlapping partitions; (b) 100-subdomain, pad-150 overlapping partitioning scheme.

### 3.3. Equivalent stress–strain reconstruction

Fig. 10 compares the macroscopic measured stress–strain response with three curves obtained from the microscale fields, each shown with a 95% confidence interval on the mean at each point.

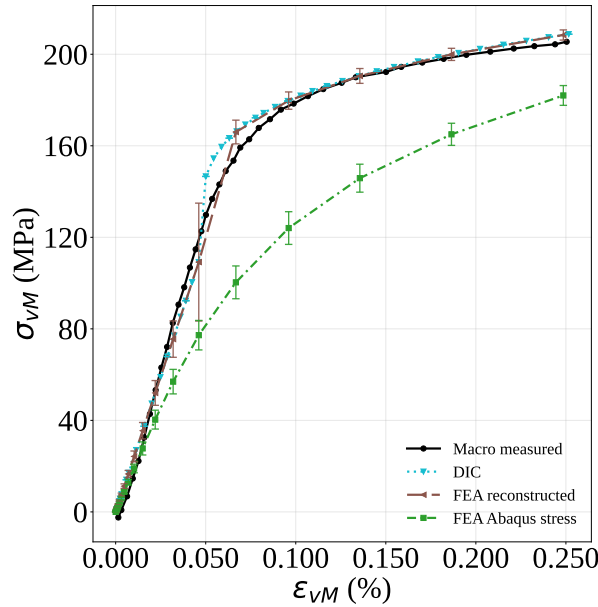
The direct finite-element stress curve is obtained by spatially averaging the Abaqus stress tensor components and applying the von Mises definition. It remains substantially lower than the macroscopic measured response throughout the plastic regime: the gap opens sharply just past the elastic–plastic transition, reaching a peak of approximately 55 MPa ( $\sim 35\%$  relative) around  $\epsilon_{vM} \approx 0.06\text{--}0.07\%$ , and only partially narrows at larger strains, remaining at roughly 10–20% even at the highest strains sampled. This is not a negligible effect. This underestimation reflects how the heterogeneous local properties and prescribed displacement boundary conditions redistribute stress within the domain: partitions with softer local parameters carry less stress than the global average would suggest, and spatial averaging of a heterogeneous stress field does not recover the globally equivalent response. The direct FE stress curve therefore cannot be compared directly to the strain-based identification used to derive the local parameters.

Two strain-based equivalent stress–strain curves are also reported, obtained by applying the scalar constitutive mapping of Section 2 ( $3G\epsilon_{vM}$  in the elastic regime, Ludwik hardening beyond yield) to the spatially averaged equivalent strains from the DIC field and from the numerical field, respectively. Both curves remain closer to the macroscopic measured response than the direct FE stress curve. The DIC-based curve rises more abruptly near the elastic–plastic transition, reflecting sensitivity to the yielding onset criterion and to local strain fluctuations in the measured field. The numerical curve follows the measured trend more progressively in this region. At larger strains, both strain-based curves converge toward the macroscopic response and reproduce the overall hardening slope with limited deviation.

These strain-based curves are not independent stress predictions: they reproject the equivalent strain through the same constitutive relation used during identification, so their agreement with the macroscopic response is primarily a consistency check on the kinematics-driven reconstruction rather than a validation of the local stress field.

The confidence interval at each point is computed from the underlying spatial (DIC) and inter-partition (FE) spread and propagated through the constitutive mapping. The interval is narrow for all three curves: for the DIC-based curve, this reflects the very large number of measurement points contributing to the spatial average, so the mean is estimated with high precision despite the material's local heterogeneity; for the two FE-based curves, it reflects the comparatively small but consistent scatter across the 25 partitions. The narrowness of the interval indicates that the small deviations from the macroscopic response are systematic rather than attributable to statistical noise in the averaging procedure.

The identification procedure remains sensitive to noise from measurement uncertainties and the regression step, particularly in pixels with limited plastic activity. The computational cost is significant: with 11.16 million elements across 25 partitions, total Abaqus run time and memory requirements scale with the number of independent stress-strain curves per partition, and are reported in Section 2.4.



**Figure 10:** Macroscopic measured stress-strain curve compared with the DIC-based, FE-based, and direct FE stress reconstructions, with 95% confidence intervals on the mean at each point.

## 4. Conclusion

The present study demonstrates that locally identified effective elastoplastic descriptors derived from full-field DIC measurements can be mapped into a pixel-resolved finite element model to reconstruct a mechanically admissible response and reproduce the dominant displacement and strain localization patterns in 316L stainless steel.

The displacement field is reproduced with very high accuracy (RMSE of  $1.32 \times 10^{-2}$  %). The equivalent strain RMSE of 0.361% reflects differences in local strain amplitudes rather than in the overall spatial organization of deformation, with the main contributions coming from stitching interface regions and from the smoothing effect of equilibrium enforcement on high-frequency DIC noise. The principal localization bands and deformation gradients are captured while the computational cost remains far below that of fully microstructure-resolved simulations. The strain-based macroscopic stress-strain reconstruction is consistent with the measured hardening trend, whereas direct averaging of finite element stress components is more sensitive to stress redistribution and post-processing choices.

The main limitation of the framework lies in highly localized regions, where strong strain gradients amplify sensitivity to measurement noise and parameter fluctuations. Equilibrium enforcement smooths these fluctuations while preserving the dominant band topology, but the sharpness and amplitude of individual localization events are underestimated.

Overall, the proposed approach provides a computationally lightweight and mechanically consistent link between full-field kinematic measurements and equilibrium-based strain reconstruction, without requiring microstructural data such as grain orientations or slip-system parameters. It is best understood as a reconstruction-oriented methodology and a practical intermediate step toward more complete microstructure-based simulations.

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## CRediT authorship contribution statement

**Adil Kılınç:** Writing – review & editing, Writing – original draft, Conceptualization, Methodology, Data curation, Software. **Qi Hu:** Writing – review & editing, Conceptualization, Methodology, Data curation. **Jean-François Witz:** Writing – review & editing, Conceptualization, Methodology, Supervision. **Denis Najjar:** Writing – review & editing, Conceptualization, Methodology, Supervision. **Vincent Magnier:** Writing – review & editing, Conceptualization, Methodology, Software, Supervision. **Ahmed El Bartali:** Writing – review & editing, Conceptualization, Methodology, Supervision, Project administration, Funding acquisition.

## 275 **A. Appendix A: Microscale Model Mapping Structure**

276 The stress–plastic strain data, based on Ludwik’s equation, are expressed as:

$$\sigma = \sigma_y + K\epsilon_p^n$$

277 For each pixel-element  $j$ , the material properties are defined as:

$$\mathbf{S}_j = (K_j, n, \sigma_j, \epsilon_p)$$

278 The mesh data consists of  $N$  nodes, each with coordinates  $(x_i, y_i)$ . To facilitate mesh-grid modifications and  
 279 accommodate different optical microscopy pixel sizes, a variable `pixel_size` is defined and set to 0.001 mm by  
 280 default. For this case, a scaling factor, 1.84, is additionally applied to the mesh coordinates:

$$(x'_i, y'_i) = (x_i \cdot \text{scale\_factor}, y_i \cdot \text{scale\_factor})$$

281 For boundary nodes, the displacement in the  $x$ - and  $y$ -directions,  $u_x$  and  $u_y$ , is interpolated using the known  
 282 displacements at two points. Given two points  $(x_1, y_1)$  and  $(x_2, y_2)$  with corresponding displacements  $u_{x,1}, u_{y,1}$  and  
 283  $u_{x,2}, u_{y,2}$ , the displacement at a node  $(x, y)$  is calculated as:

$$u_x(x, y) = \frac{x_2 - x}{x_2 - x_1} \cdot u_{x,1} + \frac{x - x_1}{x_2 - x_1} \cdot u_{x,2} \quad (8)$$

$$u_y(x, y) = \frac{x_2 - x}{x_2 - x_1} \cdot u_{y,1} + \frac{x - x_1}{x_2 - x_1} \cdot u_{y,2} \quad (9)$$

284 The Abaqus input file (`.inp`) is structured as follows:

```

285 *NODE, node_id, x_i', y_i'
286 *ELEMENT, type=CPS4 element_id, node_1, node_2, node_3, node_4
287 *Material, name=Material-IDp *Elastic E,ν *Plastic σj,εj
  *Boundary node_id, 1, 1, u_x and node_id, 2, 2, u_y

```

## 288 **B. Appendix B: Derivation of the Plane-Stress von Mises Equivalent Strain**

289 The von Mises equivalent strain is defined from the deviatoric strain tensor as

$$\epsilon_{\text{vM}} = \sqrt{\frac{2}{3} \epsilon' : \epsilon'}, \quad (10)$$

290 where

$$\epsilon' = \epsilon - \frac{1}{3} \text{tr}(\epsilon) \mathbf{I}. \quad (11)$$

291 For a two-dimensional surface measurement under plane-stress conditions, the in-plane strain components are  $\epsilon_{xx}$ ,  
 292  $\epsilon_{yy}$ , and  $\epsilon_{xy}$ . The out-of-plane strain component is not zero and is obtained from Hooke’s law as

$$\epsilon_{zz} = -\frac{\nu}{1 - \nu} (\epsilon_{xx} + \epsilon_{yy}), \quad (12)$$

293 while

$$\epsilon_{xz} = \epsilon_{yz} = 0. \quad (13)$$

294 The strain tensor is therefore written as

$$\boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_{xx} & \epsilon_{xy} & 0 \\ \epsilon_{xy} & \epsilon_{yy} & 0 \\ 0 & 0 & \epsilon_{zz} \end{bmatrix}. \quad (14)$$

295 Using the deviatoric definition above, the equivalent strain can be expressed as

$$\epsilon_{vM} = \sqrt{\frac{2}{3} \left[ (\epsilon_{xx} - \epsilon_m)^2 + (\epsilon_{yy} - \epsilon_m)^2 + (\epsilon_{zz} - \epsilon_m)^2 + 2\epsilon_{xy}^2 \right]}, \quad (15)$$

296 where

$$\epsilon_m = \frac{1}{3} (\epsilon_{xx} + \epsilon_{yy} + \epsilon_{zz}). \quad (16)$$

297 Substituting the plane-stress expression for  $\epsilon_{zz}$  and simplifying gives the plane-stress von Mises equivalent strain  
298 used in this work:

$$\epsilon_{vM}^{ps} = \frac{2}{3(1-\nu)} \sqrt{(1-\nu)^2 (\epsilon_{xx}^2 + \epsilon_{yy}^2 - \epsilon_{xx}\epsilon_{yy} + 3\epsilon_{xy}^2) + \nu (\epsilon_{xx} + \epsilon_{yy})^2}. \quad (17)$$

299 Here,  $\epsilon_{xy}$  denotes the tensorial shear strain component,

$$\epsilon_{xy} = \frac{1}{2} \left( \frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x} \right). \quad (18)$$

300 When the engineering shear strain  $\gamma_{xy} = 2\epsilon_{xy}$  is used instead, the shear contribution in the final expression transforms  
301 into

$$3(1-\nu)^2 \epsilon_{xy}^2 = \frac{3}{4} (1-\nu)^2 \gamma_{xy}^2. \quad (19)$$



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